deep se
dependable evolvable pervasive software engineering group

# Virtual office hours 2


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POLITECNICO DI MILANO

# Recurrent questions

- Streams
- Containers
- MPI

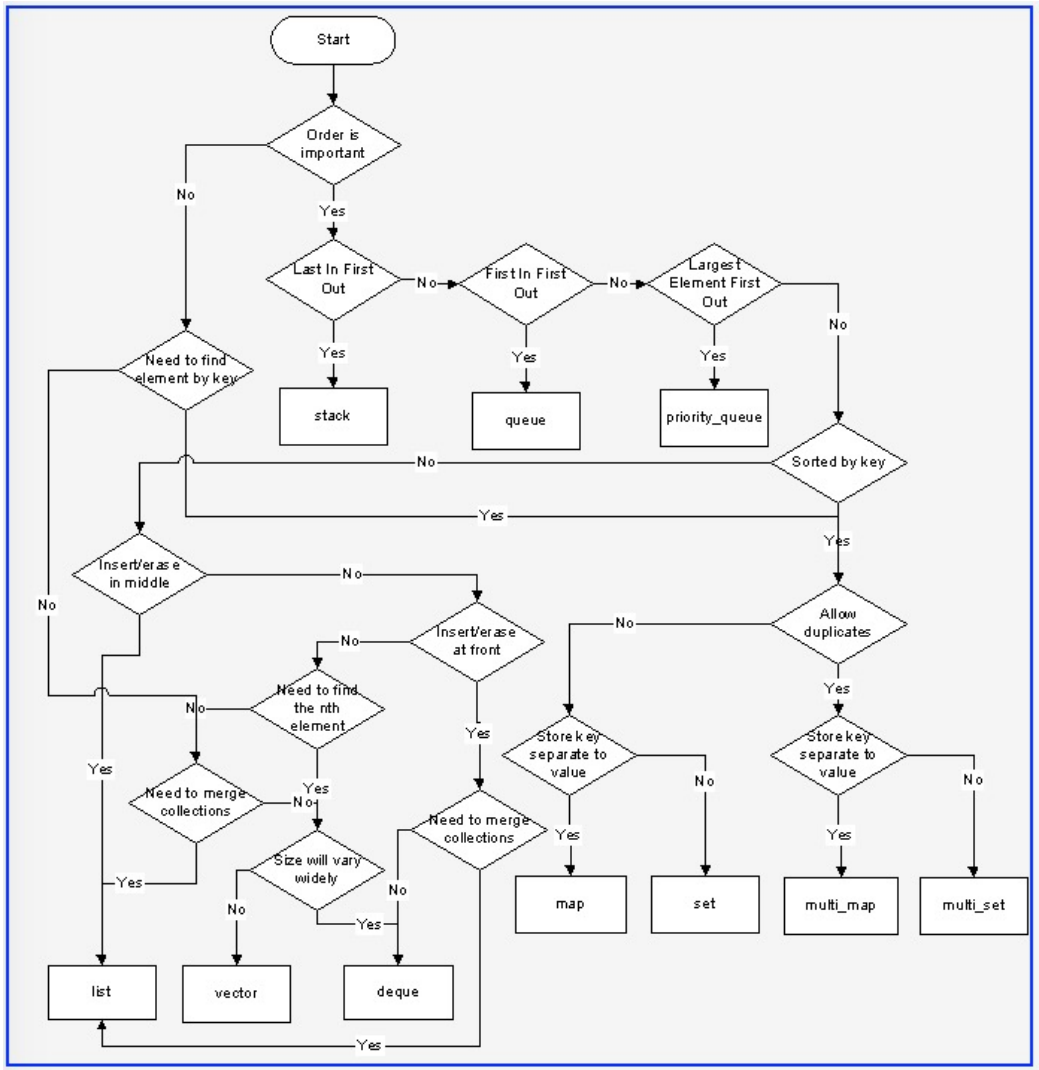
# Streams

- They are not part of exams
  - Goal: read a csv file
  - Code is always the same (and I Google if I have to implement myself!)

# Containers

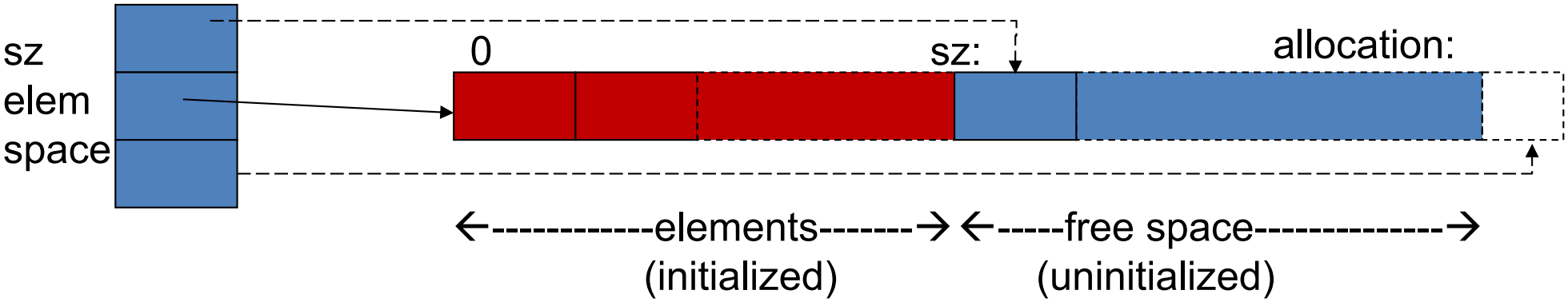
- How to choose
- Complexity of operations

# Containers



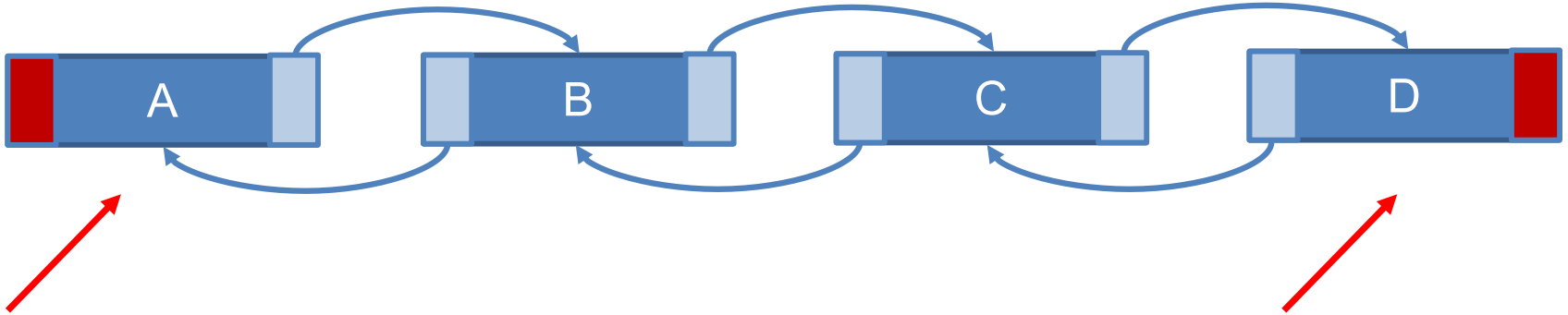

https://github.com/gibsjose/cpp-cheat-sheet/blob/master/Data%20Structures%20and%20Algorithms.md

# vector

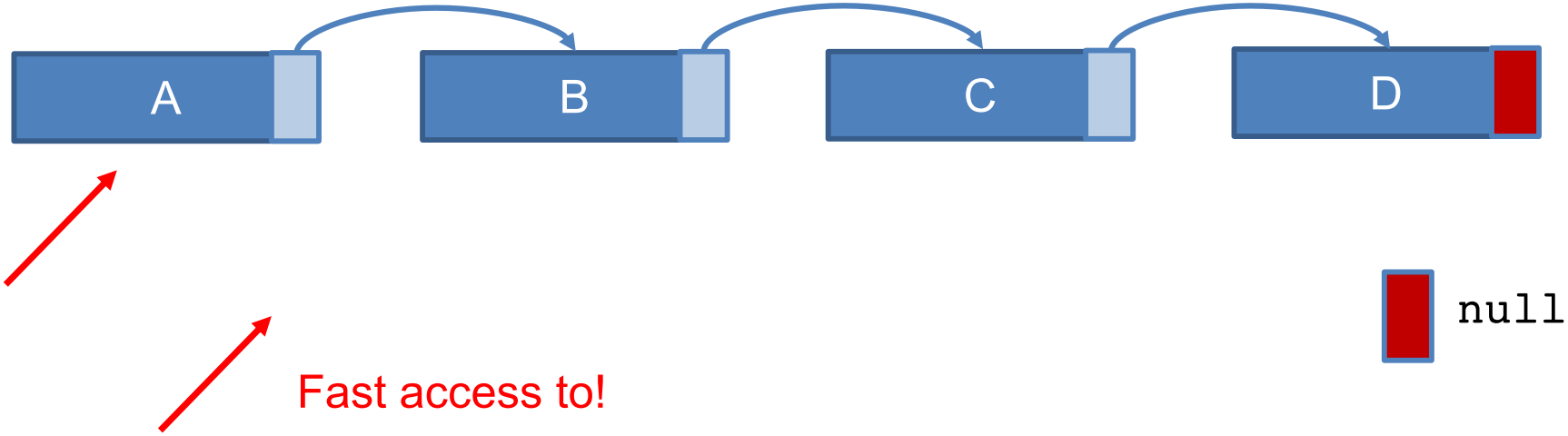

sz
elem
space
0
sz:
allocation:
←----------elements-------→
(initialized)
←-----free space----------→
(uninitialized)

# `list` and `forward_list`

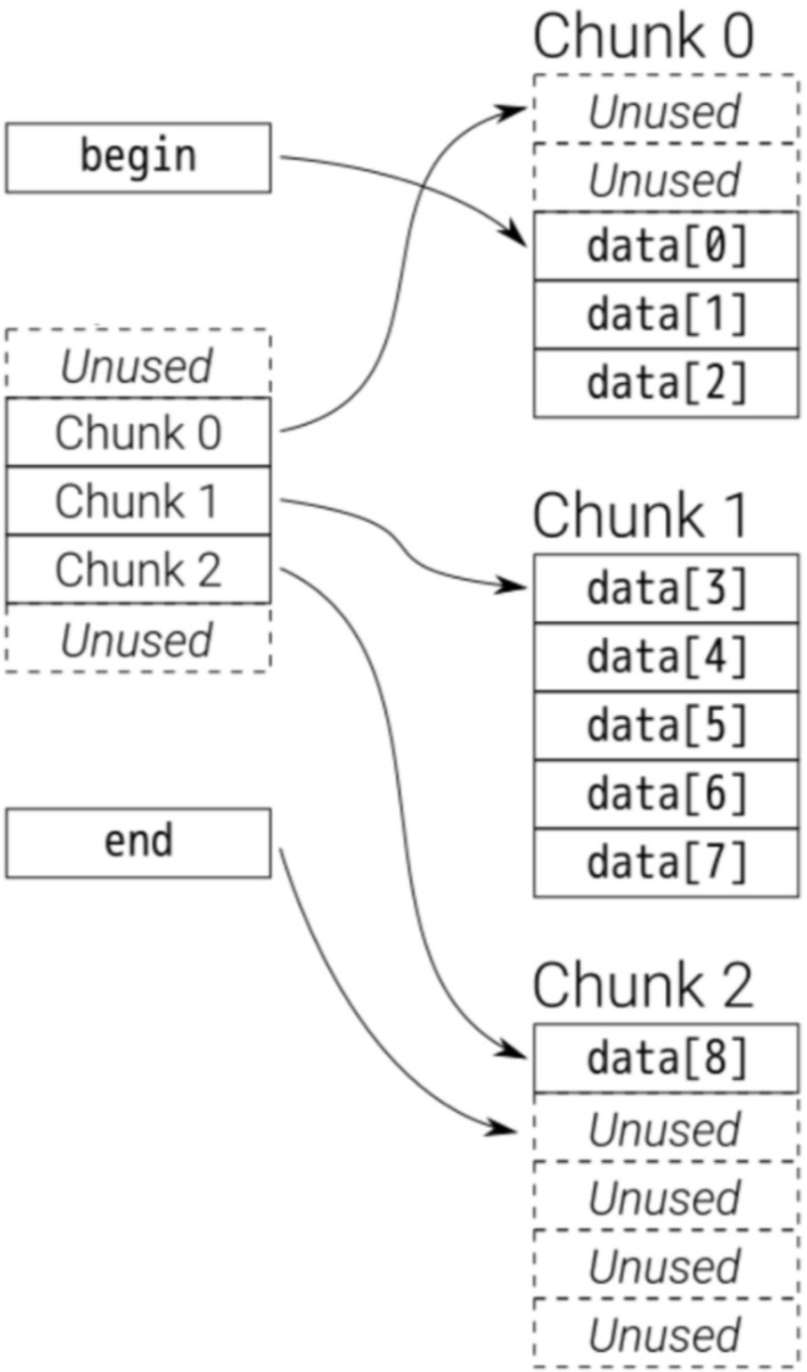
- `list` implements a doubly-linked list



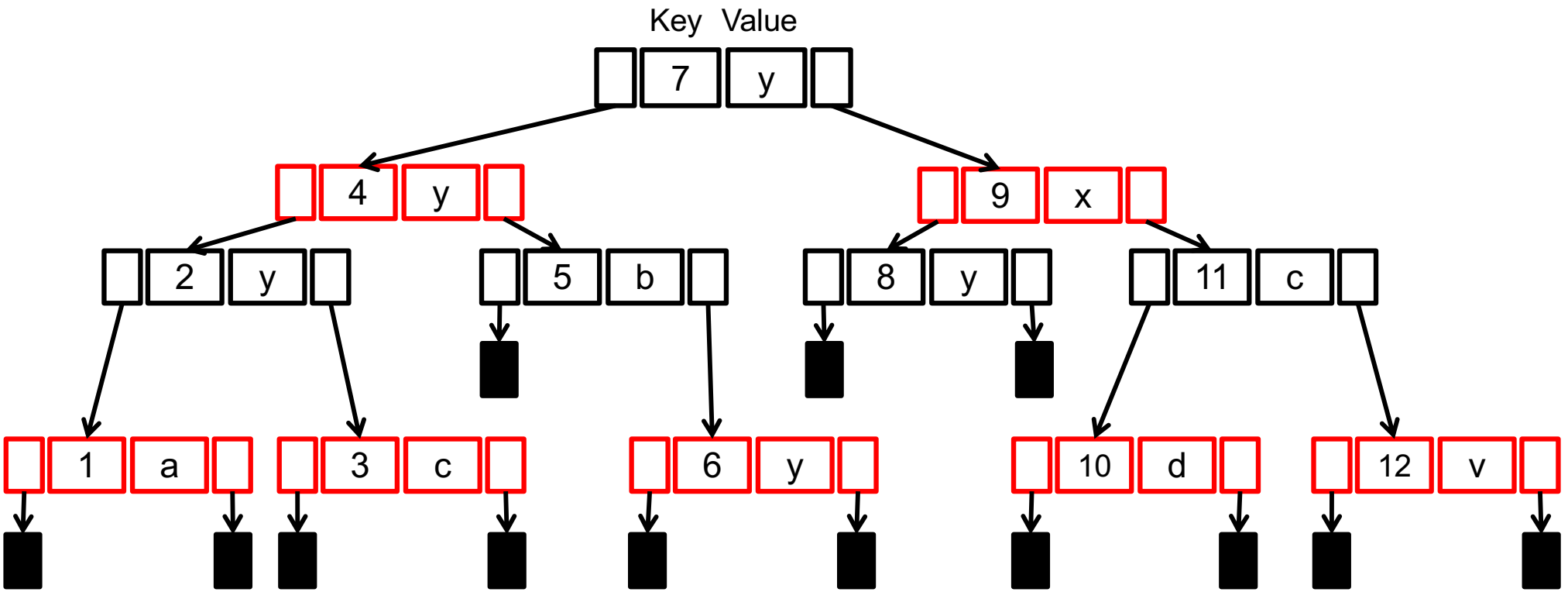

- `forward_list` implements a singly-linked list



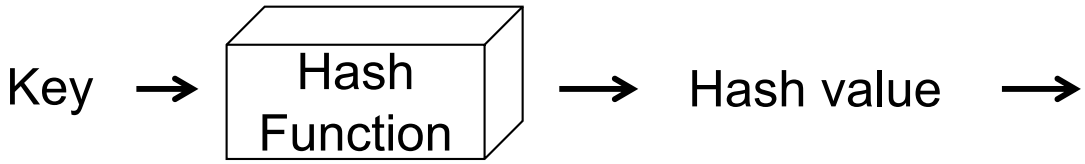
# deque


Chunk 0
Unused
Unused
data[0]
data[1]
data[2]
begin
Unused
Chunk 0
Chunk 1
Chunk 2
Unused
Chunk 1
data[3]
data[4]
data[5]
data[6]
data[7]
end
Chunk 2
data[8]
Unused
Unused
Unused
Unused

# map

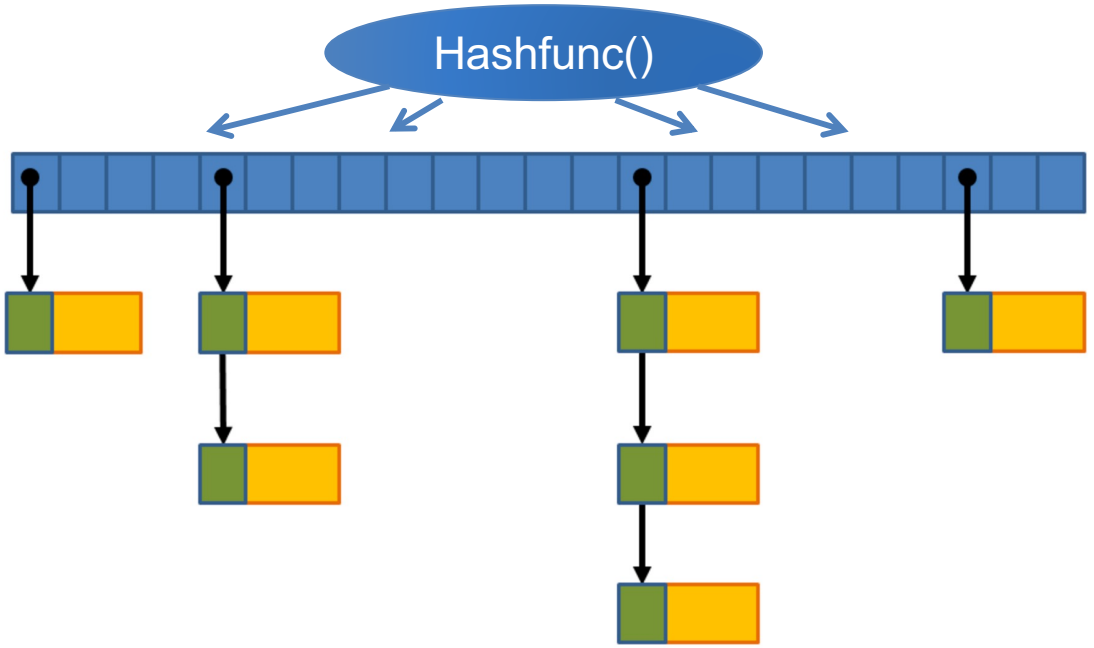


# unordered_map

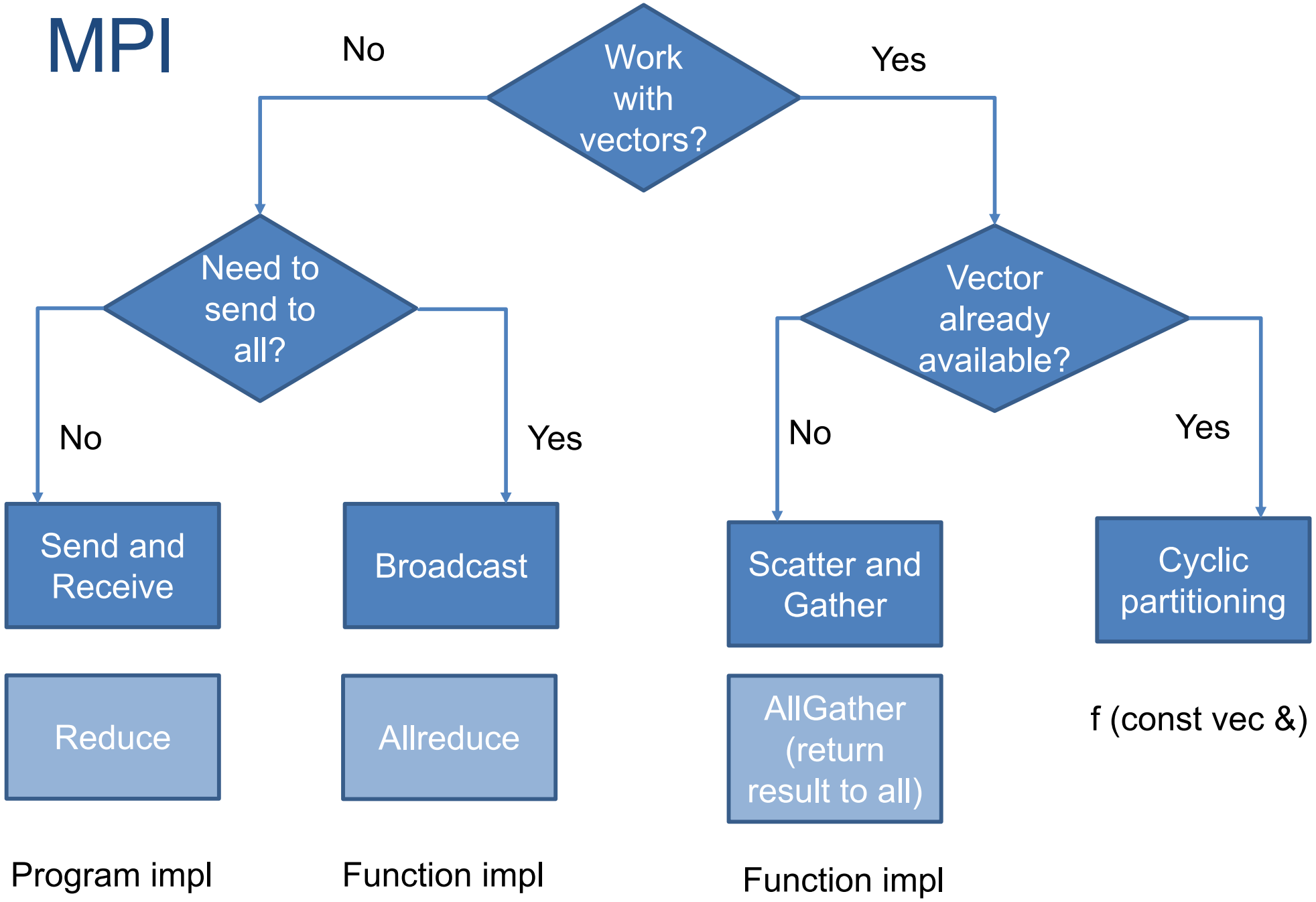



| | |
|---|---|
| **0** | 55, 85 |
| **1** | 96 |
| **2** | 37, 32 |
| **3** | |
| **4** | 104 |

buckets



# MPI


Work with vectors?
No
Yes
Need to send to all?
Vector already available?
No
Yes
No
Yes
Send and Receive
Broadcast
Scatter and Gather
Cyclic partitioning
Reduce
Allreduce
AllGather (return result to all)
f (const vec &)
Program impl
Function impl
Function impl